

Question Paper

Exam Date & Time: 26-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: CHM 1071/CHM_1071 ENGINEERING CHEMISTRY

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain the construction and working of calomel electrode as both cathode and anode. Mention any two disadvantages of it. (4)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) i) Calculate EMF of the zinc-silver cell at 25°C with $[Zn^{+2}] = 3\text{ M}$ and $[Ag^+] = 15\text{ M}$. Write the cell representation and electrode reactions. (4)
ii) Justify: Aqueous electrolytes are not preferred in lithium ion batteries.
(3+1 = 4 marks)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Give reasons for the following: (2)
i) Ambient air cannot be used in alkaline fuel cell.
ii) Pure chromium is not used as anode during chromium plating.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) i) The concentration of yeast t-RNA in an aqueous solution is 10 M. The absorbance is found to be 0.209 when this solution is placed in a 1.00 cm cuvette and 258 nm radiations are passed through it. a) Calculate the specific absorptivity (4)
b) What will be the absorbance if the solution is 5 M?
ii) With relevant graphical representation explain the conductometric titration of weak acid vs strong base.
(2+2 = 4 marks)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Define reverse osmosis. Describe with a neat diagram how sea water is purified using this technique. Write any two advantages of it. (4)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 2C) i) Define electrodialysis. (2)
ii) Distinguish between hard water and soft water

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3A) i) Describe in detail the electrochemical theory of corrosion of iron. (4)
ii) Why does a tin-coated iron rust rapidly when the tin layer is broken?

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) i) Explain the principle of sacrificial anode and impressed current methods of preventing corrosion. (4)
ii) Explain the effect of the following factors on the rate of corrosion
a) Nature of metal
b) Temperature.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3C) Write a brief note on i) Liquid metal corrosion ii) Galvanic corrosion (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4A) Explain the classification of the following. (4)
i) Particle reinforced composites
ii) Thermotropic liquid crystals based on the shapes

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4B) Explain the preparation of the following. (4)
i) Thin film by Vacuum deposition
ii) Nano materials by sol-gel method.

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 4C) i) A polymer sample whose number average molecular weight is 11200 has the following (2)
composition.
Degree of polymerization 200 300 400 500
% Composition 10 20 30 40
Calculate the molecular weight of monomer and weight average molecular weight.

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5A) i) Write the reactions involved in the electroless plating of copper. (4)
ii) The change in free energy (ΔG) for the cell $\text{Cd(s)} / \text{CdCl}_2(\text{aq}) // \text{AgCl(s)} / \text{Ag}$ at 298 K is -98.3 kJ/mol, the temperature coefficient of emf is -0.0129 V/K and the corresponding emf at 273 K is 0.8325 V. Calculate ΔH and ΔS at 273 K.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 5B) (i) Differentiate between: (4)
a) CVD and PVD.
b) Nematic and smectic phases of liquid crystals.
(ii) What is glass transition temperature? What is the influence of plasticizer on glass transition temperature?

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) Explain the intergranular corrosion observed in 18-8 stainless steel during welding. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CHM_1071:ENGINEERING CHEMISTRY

#	Course Outcome	Description	Marks
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1	1	CO 1	14
2	2	CO 2	4
3	3	CO 3	12
4	4	CO 4	6
5	5	CO 5	14
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CHM_1071:ENGINEERING CHEMISTRY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	11	34(68%)
3	Application	4	16(32%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)